Java / Spring / Microservices LinkedIn Notes Summary

1. Synchronous vs Asynchronous Programming

Synchronous (SYNC)

• Tasks execute one after another.
• Current thread waits until work finishes.
• Blocking operation.
• Easier to debug and understand.

Asynchronous (ASYNC)

• Tasks can run independently.
• Non-blocking execution.
• Better scalability and responsiveness.

Flow

```
Main Thread -----> Continue other work
        \
         ---> Background Task ---> Callback/Future
```

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2. HashMap vs ConcurrentHashMap

HashMap

• Not thread-safe
• Allows one null key and multiple null values

ConcurrentHashMap

• Thread-safe
• No null keys/values
• Better for multithreaded systems

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3. Spring Boot Annotations Cheat Sheet

Core Stereotypes

- @Component
- @Service
- @Repository
- @Controller
- @RestController

REST APIs

- @GetMapping
- @PostMapping
- @PutMapping
- @DeleteMapping

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4. Spring Framework Overview

Main Concepts

- IoC
- Dependency Injection
- AOP
- Transaction Management

Architecture

```
Application
    |
Spring Framework
 ■■■ Web
 ■■■ AOP
 ■■■ Data Access
 ■■■ Core Container
```

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5. Strategy Pattern

Benefits

• Cleaner code
• Better testing
• Easy extension
• Follows Open/Closed Principle

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6. Microservices Design Patterns

Important Patterns

• Saga Pattern
• Circuit Breaker
• Retry with Backoff
• API Gateway
• Event Driven Architecture
• Bulkhead Pattern
• Service Discovery

Flow

```
Client
   |
API Gateway
   |
-------------------------
|     |        |        |
User Order Payment Inventory
Svc  Svc   Svc      Svc
```

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7. Java Records

Example

```
record User(String name, int age) {}
```

Benefits

• Immutable
• Less boilerplate
• Auto-generated methods

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8. Java Collections Framework

Main Types

• List

• Set

• Queue

• Map

Collection Hierarchy

```
Collection
 ■■■ List
 ■■■ Set
 ■■■ Queue

Map
 ■■■ HashMap
 ■■■ TreeMap
 ■■■ ConcurrentHashMap
```

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9. Java 8 Features

Features

• Lambda Expressions

• Stream API

• Functional Interfaces

• Optional

• Date & Time API

Lambda Example

```
list.forEach(name -> System.out.println(name));
```

Final Interview Preparation Notes

Focus Areas

1. Collections

2. Java 8
3. Spring Boot
4. Microservices
5. Concurrency
6. Design Patterns

Recommended Order